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Mapping self-reported to behavioral impulsiveness: The role of task parameters

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Abstract

Despite being regarded as indicators of a common psychological capacity, behavioral and self-reported measures of impulsiveness have been found to barely correlate with each other. Acknowledging the construct's multidimensional nature, the present study set out to map dissociable components of behavioral self-control (delay discounting, response inhibition) onto lower-order facets of self-reported impulsiveness. In addition, we examined whether the relationship between response inhibition and self-reported impulsiveness depends on the balance between facilitative and interfering priming processes involved in a laboratory task. In two consecutive studies, 185 participants completed laboratory self-control tasks as well as self-report questionnaires designed to measure facets of impulsiveness. Correlational analyses revealed an association between subscales of the Barratt Impulsiveness Scale (BIS) and response inhibition in a go/no-go paradigm involving simultaneously presented task-irrelevant distractors. This association vanished when an onset asynchrony between distractor and target stimuli was introduced. Previous findings regarding correlations between BIS subscales and delay discounting or intra-individual response variability could not be replicated. Results indicate that the relationship between response inhibition and self-reported impulsiveness critically varies as a function of subtle task parameters. Focusing on these procedural details and the multidimensionality of self-reported impulsiveness might allow for a more differentiated analysis of the convergent validity of self-control measures.

Keywords: self-control, impulsiveness, convergent validity, response inhibition, interference, stimulus onset asynchrony

Mapping self-reported to behavioral impulsiveness: The role of task parameters

Concepts of self-control, self-regulation, inhibitory control, impulse control and impulsiveness have received attention from various disciplines in the behavioral sciences including behavioral psychology and behavioral economics (Rachlin & Green, 1972; Ainslie, 1975), personality psychology (Eysenck & Eysenck, 1977), developmental psychology (Dempster, 1992), cognitive science (Posner & Rothbart, 1998), social psychology (Metcalf & Mischel, 1999) and cognitive neuroscience (Cohen & Lieberman, 2010). As a consequence, experimental and theoretical models differ substantially across research groups. Behaviorist approaches, for instance, focus on operational definitions describing self-control “as the choice of a larger, more delayed reinforcer over a smaller, less delayed reinforcer” (Logue & King, 1991, p. 105) and have inspired a large body of research concerned with animal and human behavior in delay-discounting paradigms (Mazur, 1988; Kirby & Marakovic, 1996). The cognitivist concept of inhibitory control (i.e., the suppression of prepotent responses; Miyake et al., 2000), on the other hand, has motivated the development of laboratory tasks measuring participants’ ability to withhold responses (as in go/no-go tasks or stop-signal paradigms; Logan & Cowan, 1984) or their reaction time when exposed to incongruent stimulus configurations (Stroop, 1935; Eriksen & Eriksen, 1974). Further, studying individual differences within the field of personality psychology has led to the development of multiple psychometric self-report scales presumed to measure stable traits of self-control (Tangney et al., 2004) and impulsiveness (Patton, Stanford & Barratt, 1995; Whiteside & Lynam, 2001).

Dimensions of Self-Control and Impulsiveness

Different measurement traditions in the study of self-control and impulsiveness share, as heterogeneous as they might be, a common constitutive feature: they focus on the assessment of behavior involving the inhibition or modulation of prepotent responses (Eigsti

et al., 2006; Muraven & Baumeister, 2000). Hence, there is obvious appeal to the notion that self-control tasks are fueled by the same mental resource (Baumeister et al., 1998). Recently, it has been concluded that the broad empirical evidence supporting this resource model implied that “performance in different self-control tasks should be correlated with one another” (Tabibnia et al., 2011; p. 4805; see also Cohen & Lieberman, 2010). Despite its plausibility, however, this inference is likely to be oversimplified as accumulating research suggests impulsiveness to be a rather heterogeneous construct (Cyders & Coskunpinar, 2011, 2012; Duckworth & Kern, 2011; Evenden, 1999; Nigg, 2000; Sharma, Kohl, Morgan, & Clark, 2013).

Performances on go/no-go tasks and stop-signal paradigms which require the suppression of automatic responses are, for instance, only marginally correlated with the ability to delay reward measured in temporal discounting tasks (Duckworth & Kern, 2011). Hence, response inhibition and delay discounting have been proposed to be orthogonal dimensions of impulsiveness although both processes appear to involve the control of prepotent responses (Dom, De Wilde, Hulstijn & Sabbe, 2007; Reynolds, Ortengren, Richards & de Wit, 2006).

Accounting for the multifaceted nature of the construct requires both coherent theoretical frameworks (Dick et al., 2010; Dougherty, Mathias, Marsh, 2005; Nigg, 2000) and reliable empirical evidence dissociating different facets of self-control and impulsiveness. It has been suggested to distinguish between self-control processes on the basis of factor-analytical methods (Friedman & Miyake, 2004), differential predictive validity for relevant outcomes (Duckworth & Kern, 2011), neurally dissociable circuitries (Dalley et al., 2011, Nee et al., 2007), or distinct electrophysiological correlates (Brydges et al., 2012). An alternative approach is illustrated by a study by Enticott et al. (2006) who mapped performance scores on different self-control tasks on different subscales of the Barratt

Impulsiveness Scale (BIS; Patton, Stanford & Barratt, 1995), an established self-report measure of trait impulsiveness (see also Eisenberg et al., 2007; Gorlyn, Keilp, Tyron, & Mann, 2005; Keilp, Sackeim, & Mann, 2005; Mobini, Grant, Kass, & Yeomans, 2007; Swann, Bjork, Moeller, & Dougherty, 2002; see Cyders & Coskunpinar, 2011, for review).

Studies applying both behavioral and psychometric measures of self-control regularly fail to obtain significant correlations between task performances and BIS sum scores (Dom et al., 2007; Janis & Nock, 2009; Reynolds et al., 2006; Weller, Cook, Avsar & Cox, 2008; but see also Logan, Schachar & Tannock, 1997). As a consequence, average correlation coefficients reported in recent meta-analyses (Cyders & Coskunpinar, 2011; Duckworth & Kern, 2011) are negligibly small. Systematically mapping task performances on lower-order facets of self-reported impulsiveness might, however, provide more specific and fruitful insights into the complex nature of behavioral self-control and self-reported impulsiveness, respectively, as well as into the nontrivial relationship between the both. Impulsiveness as assessed by the BIS is commonly believed to be composed of three second-order factors: motor impulsiveness (i.e., acting without forethought; Im), non-planning impulsiveness (i.e., acting without consideration for future consequences; Inp) and attentional impulsiveness (i.e., having difficulty maintaining task-relevant attention; Ia) (Patton et al., 1995). In two previous studies Inp was related to the number of false alarms in a go/no-go task, a typical measure of response inhibition (Swann et al., 2002, Enticott et al., 2006). In the study of Keilp et al. (2005), however, go/no-go performance correlated with Im and Ia while being unrelated to Inp. Moreover, stop-signal reaction time, another established indicator of response inhibition, was significantly related to Im in the study of Gorlyn et al. (2005), whereas the same measure did not correlate with any BIS subscale in the study of Enticott et al. (2006). With regard to delay discounting, two studies found relatively strong relationships between individual discounting rates and Inp (Swann et al., 2002; Mobini et al., 2007). However, this association

could not be replicated in the studies by Eisenberg et al. (2007) and Dom et al. (2007), with the latter investigation also failing to find any correlations between response inhibition and BIS lower-order factors.

The Present Study

In view of this high ambiguity and the rather small sample sizes in some of the studies mentioned above (e.g., $N = 31$, Enticott et al., 2006, $N = 32$, Swann et al., 2002, $N = 40$, Gorlyn et al., 2005), future studies involving well-established self-control tasks are necessary to further elucidate the complex relationship between BIS subscales and facets of behavioral self-control. In order to take a first step towards meeting this requirement, we collected self-reported and behavioral data reflecting the self-control abilities of $N = 185$ participants in two related studies.

The goal of Study 1 was to examine the distinction between response inhibition and delay discounting suggested by the findings of Reynolds et al. (2006) and Dom et al. (2007). Specifically, we intended to investigate whether these dimensions of behavioral impulsiveness are differentially related to one or multiple of the BIS second-order facets. Since previous studies have generated contradictory evidence with regard to this question, we aimed to contribute to a less ambiguous picture by including a larger sample of participants. Study 2 was designed to replicate and scrutinize our findings obtained from Study 1. By making a subtle change in task parameters (i.e., introducing a stimulus-onset asynchrony (SOA) between target stimuli and task-irrelevant distractors), we attempted to alter the balance between facilitative and interfering priming processes (Flowers, 1990). While interference from incongruent distractors arises both when target and distractors are presented simultaneously and when they are separated by a small temporal gap, congruent distractors facilitate responding only when presented prior to target onset (Flowers, 1990; Flowers & Wilcox, 1982). This observation suggests that facilitation and interference involve different

cognitive processing mechanisms with dissociable time courses. In our first study, participants' performance on the response inhibition task could only be affected by interference effects, as targets and distractors were presented simultaneously. In our second study, introducing a short SOA allowed facilitation effects to influence behavioral performance as well. Participants still had to inhibit the interference from incongruent distractors, but they could also benefit from the facilitative effects of congruent distractors. Importantly, deficits in inhibiting interfering information have been linked to impulsiveness (Nigg, 2000), whereas the ability to benefit from facilitation should not vary as a function of individuals' impulsiveness. As a consequence, manipulating the SOA (and thus the relative amount of facilitative versus interfering effects) can be expected to modulate the association between measures of response inhibition and BIS subscale scores. Comparing the results from our two studies allowed us to evaluate whether the correlation between lower-order facets of self-reported impulsiveness and response inhibition varies according to the degree to which dissociable mechanisms of attentional information processing (facilitation, interference) are involved in the task. Finding the relationship between laboratory and self-report measures to depend on a simple task setting (i.e., the SOA) could provide a new angle on the ambiguous pattern of results generated in this area.

General Methods

Both studies were conducted in a quiet and constantly illuminated room with the experimenter being present but separated from the participant by means of a wall in the middle of the laboratory. Participants had normal or corrected-to-normal vision (with red-green color deficiency being screened out) and received course credit which was not contingent on performance. Experimental tasks were designed using OpenSesame 0.24 (Mathôt, Schreij, & Theeuwes, 2012) and presented on a 24 inch flat screen (Samsung, Seoul, South Korea).

In order to assess self-reported impulsiveness, the 11th version of the BIS (BIS-11, Patton et al., 1995) was administered in both studies. On the 30 items of the BIS, participants are asked to indicate how frequently they engage in impulsiveness-related behaviors and cognitions on a four-point scale (from 1 = rarely/never to 4 = almost always/always). The BIS-11 has been demonstrated to have good internal consistency (Patton et al., 1995) and the German translation used in the present study (Preuss et al., 2007) proved to be reliable as well ($\alpha = .87$). Factor analysis identified three second-order factors which were, in turn, composed of six first-order factors (Patton et al., 1995). To generate comparable results, we restricted our analyses to the second-order facets attentional impulsiveness (Ia), motor impulsiveness (Im) and non-planning impulsiveness (Inp) (Enticott et al., 2006; Gorlyn et al., 2005; Keilp et al., 2005; Mobini et al., 2007; Swann et al., 2002).

Statistical analyses were performed using SPSS 21.0 (IBM, Armonk, NY). The level of significance was set at $\alpha = .05$. To control for multiple testing, alpha levels were adjusted by Bonferroni correction. Alpha levels were divided by nine in Study 1 (examining correlations between three behavioral measures and three BIS subscales) and by six in Study 2 (examining correlations between two behavioral measures and three BIS subscales). Correlation coefficients were compared according to the procedure outlined by Meng, Rosenthal, and Rubin (1992). Using G*Power 3.1.5 (Faul, Erdfelder, Lang, & Buchner, 2007), we conducted sensitivity analyses identifying the smallest effect sizes detectable with sufficient power ($1 - \beta = .80$) given the sample size of the respective study.

Study 1

Tasks and Procedure

After receiving general instructions and providing informed consent, $N = 70$ participants (62 female, mean age = 21.8 years, $SD = 2.6$ years) were tested on a hypothetical money-choice task assessing delay-discounting rates. Before they engaged in the second

laboratory task, a go/no-go paradigm designed to measure response inhibition, participants were asked to complete the study questionnaires including the BIS-11.

Delay discounting (DD). Individuals' tendency to discount delayed rewards was assessed by means of a computerized choice procedure forcing participants to state a preference between two simultaneously displayed monetary rewards (Rachlin, Raineri & Cross, 1991). Across 184 trials, participants hypothetically chose between a large delayed reward (fixed amount of 1000€ ['A' in equation (1)], variable delay ['D' in equation (1)]) and a smaller immediate reward (variable amount, no delay). Delays of the fixed amount of money were varied from shortest to longest (4 weeks, 2 months, 3 months, 4 months, 5 months, half a year, 9 months, 12 months). Within each delay condition, the immediate options were presented in ascending order (100€, 150€, 200€, 250€, 300€, 350€, 400€, 450€, 500€, 550€, 600€, 650€, 700€, 750€, 800€, 850€, 900€, 920€, 940€, 960€, 980€, 990€, 1000€). For each of these delay conditions, indifference points (i.e., the subjective values of the delayed reward, 'V' in equation (1)) were determined as the average between the largest rejected and the smallest accepted immediate reward. Fitting Mazur's formula of hyperbolic delay discounting (Mazur, 1988)

$$V = \frac{A}{(1 + kD)} \quad (1)$$

to these indifference points using a nonlinear curve-fitting program (OriginPro 8.6, OriginLab Corporation, Northampton, MA, USA) yielded estimates of individual discounting rates (k). In order to normalize the resulting sample distribution, k values were natural-log transformed. Since seven participants did never prefer the smaller immediate reward over the larger delayed reward ($k = 0$), a constant (0.0001) had to be added to individual discounting parameters to enable log-transformation.

[insert Figure 1]

Go/no-go task (GNG). In order to quantify participants' ability to inhibit inappropriate responses, a classic go/no-go task was designed (Eigsti et al., 2006; Lange & Eggert, 2014; Lange, Seer, Rapior, Rose, & Eggert, 2014; see Figure 1). Participants were required to respond (press the space bar) as fast as possible whenever a frequent go stimulus (red circle, frequency: 80%) appeared in the center of the computer screen, but to withhold responses to infrequent no-go stimuli (green circle, frequency: 20%) for 1500 ms. Establishing a presentation ratio (go vs. no-go) of 4:1 and an "unnatural" stimulus-response mapping (red = go; green = no-go) forced participants to inhibit prepotent go responses when confronted with an infrequent no-go stimulus. During 75 initial training trials, participants instrumentally learned about task contingencies via visual and auditory feedback. During 450 subsequent test trials, only negative auditory feedback was provided after misses (not responding to a go stimulus within 1500 ms) and false alarms (responding to a no-go stimulus). In order to increase the demand for inhibitory control, four task-irrelevant distractors (circles randomly varying in color) were configured around the central target stimulus. The frequency of red and green distractor stimuli was 11%, respectively, with the remaining 78% of stimuli being presented in colors varying on a wavelength gradient between red and green. Between trials, a fixation cross was presented for 600 ms (inter-trial interval, ITI) in the center of the display. The number of false alarms (or errors of commission; Reynolds et al., 2006) served as behavioral measure of response inhibition. In addition, intra-individual response variability was quantified by calculating the coefficient of variability (COV; $(SD/mean) \times 100$) for response latencies to go stimuli. Low performance variability in RT tasks has been argued to index efficient top-down executive control (Bellgrove, Hester, & Garavan, 2004; Stuss, Murphy, Binns, & Alexander, 2003). Importantly, Enticott et al., (2006) reported an association between the COV in a go/no-go

task and the BIS subscale Im. As this finding relies on a rather small sample size ($N = 31$), we aimed to investigate whether it can be replicated in a larger sample of participants.

[insert Table 1]

Results and Discussion

Summary statistics of task performances as well as Pearson correlation coefficients are displayed in Table 1. BIS subscales had acceptable internal consistency (Ia: $\alpha = .72$, Im: $\alpha = .76$, Inp: $\alpha = .73$), with Cronbach's alpha for the BIS sum score being .87. Sensitivity analysis revealed that correlations between measures of impulsiveness of at least .29 could have been detected with sufficient power ($1 - \beta = .80$). Correlation coefficients had to be as large as .31 and .34 to be detected with 85% and 90% power, respectively.

In line with the findings by Reynolds et al. (2006) and Dom et al. (2007), the correlation between the number of false alarms in the go/no-go task and participants' delay-discounting rates was small and not significant ($r = .12$, $p = .31$). Hence, our data support the idea that response inhibition and temporal discounting are independent dimensions of behavioral impulsiveness.

Although the present study was powerful enough to detect medium-to-large correlations, we could not replicate the associations between Inp and delay-discounting performance reported by Swann et al. (2002) and Mobini et al. (2007). Adding to the negative findings reported by Dom et al. (2007) and Reynolds et al. (2006), our results indicate that self-reported impulsiveness and measures of temporal discounting do not share a significant amount of common variance.

While delay-discounting performance was not related to any of the BIS subscales, the number of failures to inhibit responses to no-go stimuli was significantly associated with Ia ($r = .41$, $p < .001$) and Im ($r = .27$, $p < .05$), but not with Inp ($r = .23$, $p = .05$). When corrected for multiple testing (adjusted $\alpha = .0056$), the correlation between response inhibition and Ia

remained significant, while the correlation between response inhibition and Im did not. These results mirror the findings of Keilp et al. (2005), but are in stark contrast with the studies by Swann et al. (2002) and Enticott et al. (2006) who obtained the reversed pattern with only Inp being related to participants' no-go performance. This apparent contradiction might be attributed to a high level of task heterogeneity between studies. Go/no-go tasks are rarely administered in the exact same way by different research groups and slight changes in task parameters might cause cognitive processes on different levels of information processing to be involved in task performance (Nee et al., 2007). This hypothesis will be further investigated in Study 2.

Importantly, the magnitude of the correlation between Ia and the number of false alarms does not significantly differ from either of the other two correlations (false alarms and Im: $z = 1.42$, $p = .15$; false alarms and Inp: $z = 1.20$, $p = .23$). Hence, the association between behavioral and self-reported impulsiveness can be regarded to be rather unspecific, as response inhibition cannot be mapped onto a distinct BIS factor.

With regard to response variability, we did not find the correlation between COV and Im reported by Enticott et al. (2006) ($r = .15$, $p = .21$). Since our sample size was more than twice as large as the one examined by Enticott et al., our results suggest that the previously reported correlation coefficient ($r = .35$) might have been overestimated. Alternatively, it remains possible that relationships such as the one between response variability and Im crucially depend on the population they are investigated in (i.e., that there is meaningful between-population heterogeneity in the size of the correlation coefficients). Further replications are required to obtain a more realistic estimate of the association between intra-individual performance variability and self-reported impulsiveness (see Study 2).

[insert Table 2]

Study 2

Our second study focused more narrowly on the relationship between facets of self-reported impulsiveness and response inhibition demonstrated in Study 1. By assessing a measure of response inhibition almost identical to the one applied in Study 1, it was possible to examine whether our results regarding a link between Im, Ia and no-go performance could be replicated in a larger sample. In contrast to Study 1, a short delay was introduced between the onset of the task-irrelevant distractors and the task-relevant target stimuli to increase the relative amount of facilitative versus interfering distractor effects (Flowers, 1990; Flowers & Wilcox, 1982). As deficits in inhibiting interference but not alterations in benefiting from facilitation are conceptually related to impulsiveness (Nigg, 2000), the relationship between behavioral and self-report measures is likely to be attenuated when distractor stimuli are presented prior to target onset.

Task and Procedure

$N = 115$ participants (103 female, mean age = 21.8 years, $SD = 3.4$ years) were given general instructions and provided informed consent before being tested on a go/no-go task. We used the same paradigm as described above with the exception that target stimuli always appeared with a delay of 100 ms (SOA) after the onset of the task-irrelevant distractors (see Figure 1). Along the lines of Study 1, the number of false alarms served as a behavioral measure of response inhibition and COV was determined for responses to go trials. After having completed 75 training and 450 test trials, participants responded to the study questionnaires including the BIS.

Results and Discussion

Summary statistics of task performances as well as Pearson correlation coefficients are displayed in Table 2. Cronbach's alpha was .52, .68, .65 and .82 for Ia, Im, Inp and the BIS sum score, respectively. Sensitivity analysis revealed that correlations between measures

of impulsiveness of at least .23 could have been detected with sufficient power ($1 - \beta = .80$). Correlation coefficients had to be as large as .25 and .27 to be detected with 85% and 90% power, respectively.

Neither the global BIS score nor any of the respective subscales was significantly associated with the number of false alarms committed on the go/no-go task (all $R^2 < .02$). Hence, Study 2 failed to replicate the significant associations between response inhibition (i.e., the number of false alarms) and I_m and I_a observed in Study 1. As the probability to detect correlations as large as the one between response inhibition and I_a reported in Study 1 ($r = .41$) exceeds .99, this disparity is unlikely to result from random fluctuations. Instead, the lack of a significant association may be due to a subtle change in task parameters. As mentioned above, an SOA of 100 ms was introduced to alter the balance of facilitative and interfering processes of attentional priming (Flowers, 1990). In comparison to Study 1, participants committed less false alarms in Study 2 ($t(183) = 2.36, p < .05$), indicating that they indeed benefited from increased facilitation effects following the introduction of a short SOA. Response times to go stimuli also decreased as a function of SOA (Study 1: $M = 438$ ms, $SD = 79$ ms; Study 2: $M = 423$ ms, $SD = 51$ ms), although not to a significant extent ($t(183) = 1.38, p = .17$). The increased effect of facilitative processes might have obscured the relationship between response inhibition and BIS scores observed in Study 1. In other words, the association between response inhibition and self-reported impulsiveness appears to be restricted to situations characterized by a relative overweight of interfering versus facilitative distractor effects. However, it has to be noted that BIS subscale scores showed surprisingly low internal consistency in Study 2. We find it difficult to attribute this divergence from Study 1 to anything but random variance, as Study 2 only differed with regard to the SOA and the absence of a delay discounting task. More crucially, however, we believe that the absence of significant correlations is rather unlikely to be due to a lack of reliability, as the

BIS sum score (being significantly related to response inhibition in Study 1) had good internal consistency in Study 2, but did not correlate with no-go performance. Note that the statistical power for this analysis (i.e., the probability to detect a correlation as large as in Study 1 ($r = .36$) within a sample of 115 individuals) amounted to 99% as well. If future studies could confirm that the relationship between laboratory and self-reported measures of impulsiveness depends on a simple experimental parameter like the length of the SOA, an important step could be made towards a better understanding of the contradictory literature in this domain.

In addition to the main finding of Study 2 presented above, we report a second failed attempt to replicate the association between response variability (COV) and Im proposed by the study of Enticott et al. (2006). As the sample was larger and the correlation even weaker than in Study 1, the significant correlation reported by Enticott et al. is very likely to be either a false positive or highly population-specific.

General Discussion

In two consecutive studies, we attempted to map individuals' performance on behavioral tasks of delay discounting and response inhibition onto distinct facets of self-reported impulsiveness. We report an association between BIS subscales and response inhibition in a go/no-go task which is limited to situations a relative overweight of interfering versus facilitative distractor effects.

The Complex Relationship between Behavioral and Self-Reported Impulsiveness

Were all the measures of impulsiveness examined in our studies (delay discounting parameter, false alarms to no-go stimuli, response variability, BIS scores) conceived to be indicators of a common personality trait, our results would have to be interpreted as suggesting rather poor convergent validity. This finding would only replicate previous failures to find substantial correlations between behavioral and self-reported impulsiveness

(see Duckworth & Kern, 2011, for an overview). As the related constructs of self-control (Duckworth & Kern, 2011), impulsiveness (Evenden, 1999; Nigg, 2000) and inhibitory control (Friedman & Miyake, 2004) have already been shown to be everything but homogeneous, finding only few reliable associations between highly specific behavioral tasks and highly aggregated self-report measures is of little surprise. In line with previous approaches (Cyders & Coskunpinar, 2011), we decided to focus instead on mapping dissociable components of behavioral self-control on lower-order factors of self-reported impulsiveness.

A recent meta-analysis dedicated to the same research question found a significant, albeit small, overlap between some laboratory measures and specific self-report facets of impulsiveness (Cyders & Coskunpinar, 2011). Across studies, the reported lack of planning (a dimension of impulsiveness proposed by Whiteside & Lynam, 2001) was, for instance, reliably associated with delay-discounting performance ($r = .13$) and response inhibition ($r = .10$), while the magnitude of these correlation coefficients might have been inflated by the file drawer problem. In view of this evidence, it could be argued that any associations between behavioral and self-reported measures of impulsiveness are negligibly small and that these measures are mostly tapping into disparate constructs (Cyders & Coskunpinar, 2011). However, we believe that our data provide at least two insights that are incremental to these sobering conclusions about the relationship between lab task and self-report impulsiveness.

First, it might be necessary to adopt an even more specific focus when analyzing dimensions of self-reported impulsiveness. Following the classification of Cyders and Coskunpinar (2011) and Dick et al. (2010), Ia, Im, and Inp are considered to load on the same dimension of impulsiveness (lack of planning). This implies that, for the purpose of the meta-analysis, items like “I squirm at plays or lectures” and “I plan for job security” are treated as indicators of the same facet of impulsiveness. When relating this facet to a behavioral

measure reflecting the behavior in response to a highly specific (and artificial) stimulus conflict, low correlations have to be expected. We do not claim that the BIS factor structure mirrors the heterogeneity of impulsiveness facets in an ideal way. However, this approach allowed us, for instance, to identify a rather substantial association between attentional impulsiveness and response inhibition ($r = .41$). For the future, we concur with Cyders and Coskunpinar (2011) that research in this area should rely on factor-analytical studies assessing multiple measures of behavioral (Miyake et al., 2000) and self-reported (Whiteside & Lynam, 2001) impulsiveness.

Second, we were able to demonstrate that the significant associations between response inhibition and BIS subscale scores observed in Study 1 vanished when an SOA was introduced to the otherwise unchanged task (Study 2). Most likely, this disparity can be attributed to the differential involvement of cognitive processes at an earlier stage of information processing (Friedman & Miyake, 2004; Nee et al., 2007). Manipulating the onset asynchrony of target and distractor stimuli allows analyzing cognitive processes with different time courses (Flowers & Wilcox, 1982). Incongruent distractor stimuli interfere with target detection both at short and absent SOAs, whereas the facilitative priming effects of congruent distractors are restricted to situations involving a short SOA. If the correlation between the number of false alarms and self-reported impulsiveness disappears with the introduction of an SOA, the altered task can be concluded to recruit cognitive mechanisms which are unrelated to the trait(s) measured by the BIS scales. Our results suggest that response inhibition under *distractor interference only* relates to self-reported impulsiveness (especially to Ia), while there is no such association when individuals have to inhibit responses to no-go stimuli under distractor interference *and* distractor facilitation. Allowing facilitative priming effects to influence participants' no-go performance thus appears to

significantly attenuate the extent to which response inhibition tasks and self-report scales of impulsiveness tap into the same construct.

Understanding the relationship between performance scores and self-report measures as being sensitive to variations of experimental parameters can also account for the divergent results obtained in previous studies (Swann et al., 2002; Keilp et al., 2005; Gorlyn et al., 2005; Enticott et al., 2006). Future research on the convergence of behavioral and self-reported self-control measures could most likely benefit from focusing on experimental features (e.g., the SOA, the stimulus material involved, instructions encouraging specific speed-accuracy trade-offs) as a crucial source of variance. It is possible that we have tried to answer the wrong research question in asking “Are behavioral and self-reported measures of self-control measuring the same construct?” Instead, a novel research program might be needed to examine under which conditions a specific behavioral measure is related to a specific facet of psychometric impulsiveness.

Limitations

Some caveats warrant mention when interpreting the current results. Since the sample consisted exclusively of undergraduate psychology students, findings cannot be generalized beyond this selective population of predominantly female and highly educated individuals. In addition, power analysis revealed that only effects of at least medium size were likely to be detected given our sample sizes. Note however that in each of our studies more participants were tested than in any of the previous attempts to map task performances on BIS subscales. Moreover, we reasoned that finding statistically significant associations between impulsiveness measures as small as meta-analytically shown (Cyders & Coskunpinar, 2011; Duckworth & Kern, 2011) would be insufficient to demonstrate convergent validity.

It has to be noted that, for the present analysis, we focused on the evaluation of only one self-report questionnaire for the assessment of impulsiveness (i.e., the BIS). Further

research on task-parameter-specific associations between behavioral and self-reported impulsiveness might benefit from including multiple self-report instruments such as the UPPS impulsive behavior scale (Whiteside & Lynam, 2001), Dickman's functional and dysfunctional impulsivity scales (Dickman, 1990), the Brief Self-Control Scale (Tangney et al., 2004) or the respective scales of Cloninger's (1994) Temperament and Character Inventory. Along similar lines, studying the effect of SOA manipulations in a go/no-go paradigm can only constitute the beginning of a systematic analysis of the relevance of task settings for questionnaire-behavior associations. Altering task demands by introducing varying levels of perceptual uncertainty, attentional conflict or response complexity, for instance, might reveal to which extent the relationship between behavioral and self-reported impulsiveness relies on dissociable inhibitory processes at different stages of information processing (Nee et al., 2007). In this context, it might be most fruitful to investigate the influence of task differences within a systematic framework such as the dimensional overlap model (Kornblum, Hasbroucq, & Osman, 1990).

Conclusion

Correlations between tasks and scales of self-control and impulsiveness are small at most. While this common observation (Duckworth & Kern, 2011) indicates that the construct's complexity has been oversimplified or underestimated in the past (Friedman & Miyake, 2004), it does not necessarily put a sobering end to the research on the convergence of self-control measures. Instead, systematically focusing on specific facets of self-reported impulsiveness and the procedural details of laboratory tasks might pave the way towards a more comprehensive understanding of the construct of self-control.

References

- Ainslie, G. (1975). Specious reward: a behavioral theory of impulsiveness and impulse control. *Psychological Bulletin*, 82(4), 463-496. doi: 10.1037/h0076860
- Baumeister, R. F., Bratslavsky, E., Muraven, M., & Tice, D. M. (1998). Ego depletion: is the active self a limited resource? *Journal of Personality and Social Psychology*, 74(5), 1252-1265. doi: 10.1037/0022-3514.74.5.1252
- Bellgrove, M. A., Hester, R., & Garavan, H. (2004). The functional neuroanatomical correlates of response variability: evidence from a response inhibition task. *Neuropsychologia*, 42(14), 1910-1916. doi: 10.1016/j.neuropsychologia.2004.05.007
- Brydges, C. R., Clunies-Ross, K., Clohessy, M., Lo, Z. L., Nguyen, A., Rousset, C., ... & Fox, A. M. (2012). Dissociable Components of Cognitive Control: An Event-Related Potential (ERP) Study of Response Inhibition and Interference Suppression. *PloS one*, 7(3), e34482. doi: 10.1371/journal.pone.0034482
- Cloninger, C. R. (1994). *The Temperament and Character Inventory (TCI): A guide to its development and use*. Washington University, St. Louis, MI: Center for the Psychobiology of Personality.
- Cohen, J. R., & Lieberman, M. D. (2010). The common neural basis of exerting self control in multiple domains. In: Y. Trope, R. Hassin, K. N. Ochsner (Eds.), *Self Control in Society, Mind, and Brain* (pp. 141-162). New York, NY: Oxford University Press. doi: 10.1093/acprof:oso/9780195391381.003.0008
- Cyders, M. A., & Coskunpinar, A. (2011). Measurement of constructs using self-report and behavioral lab tasks: Is there overlap in nomothetic span and construct representation for impulsivity? *Clinical Psychology Review*, 31(6), 965-982. doi: 10.1016/j.jrp.2011.11.005

- Cyders, M. A., & Coskunpinar, A. (2012). The relationship between self-report and lab task conceptualizations of impulsivity. *Journal of Research in Personality*, 46(1), 121-124. doi: 10.1016/j.jrp.2011.11.005
- Dalley, J. W., Everitt, B. J., & Robbins, T. W. (2011). Impulsivity, compulsivity, and top-down cognitive control. *Neuron*, 69(4), 680-694. doi: 10.1016/j.neuron.2011.01.020
- Dempster, F. N. (1992). The rise and fall of the inhibitory mechanism: Toward a unified theory of cognitive development and aging. *Developmental Review*, 12(1), 45-75. doi: 10.1016/0273-2297(92)90003-K
- Dick, D. M., Smith, G., Olausson, P., Mitchell, S. H., Leeman, R. F., O'Malley, S. S., & Sher, K. (2010). Review: understanding the construct of impulsivity and its relationship to alcohol use disorders. *Addiction Biology*, 15(2), 217-226. doi: 10.1111/j.1369-1600.2009.00190.x
- Dickman, S. J. (1990). Functional and dysfunctional impulsivity: personality and cognitive correlates. *Journal of Personality and Social Psychology*, 58(1), 95-102. doi: 10.1037/0022-3514.58.1.95
- Dom, G., De Wilde, B., Hulstijn, W., & Sabbe, B. (2007). Dimensions of impulsive behaviour in abstinent alcoholics. *Personality and Individual Differences*, 42(3), 465-476. doi: 10.1016/j.paid.2006.08.007
- Dougherty, D. M., Mathias, C. W., Marsh, D. M., & Jagar, A. A. (2005). Laboratory behavioral measures of impulsivity. *Behavior Research Methods*, 37(1), 82-90. doi: 10.3758/BF03206401
- Duckworth, A. L., & Kern, M. L. (2011). A meta-analysis of the convergent validity of self-control measures. *Journal of Research in Personality*, 45(3), 259-268. doi: 10.1016/j.jrp.2011.02.004

- Eigsti, I. M., Zayas, V., Mischel, W., Shoda, Y., Ayduk, O., Dadlani, M. B., ... & Casey, B. J. (2006). Predicting cognitive control from preschool to late adolescence and young adulthood. *Psychological Science*, 17(6), 478-484. doi: 10.1111/j.1467-9280.2006.01732.x
- Eisenberg, D. T., MacKillop, J., Modi, M., Beauchemin, J., Dang, D., Lisman, S. A., ... & Wilson, D. S. (2007). Examining impulsivity as an endophenotype using a behavioral approach: a DRD2 TaqI A and DRD4 48-bp VNTR association study. *Behavioral and Brain Functions*, 3: 2. doi: 10.1186/1744-9081-3-2
- Enticott, P. G., Ogloff, J. R., & Bradshaw, J. L. (2006). Associations between laboratory measures of executive inhibitory control and self-reported impulsivity. *Personality and Individual Differences*, 41(2), 285-294. doi: 10.1016/j.paid.2006.01.011
- Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Attention, Perception, & Psychophysics*, 16(1), 143-149. doi: 10.3758/BF03203267
- Evenden, J. L. (1999). Varieties of impulsivity. *Psychopharmacology*, 146(4), 348-361. doi: 10.1007/PL00005481
- Eysenck, S. B., & Eysenck, H. J. (1977). The place of impulsiveness in a dimensional system of personality description. *British Journal of Social and Clinical Psychology*, 16(1), 57-68. doi: 10.1111/j.2044-8260.1977.tb01003.x
- Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39, 175-191. doi: 10.3758/BF03193146
- Flowers, J. H. (1990). Priming effects in perceptual classification. *Perception & Psychophysics*, 47(2), 135-148. doi: 10.3758/BF03205978

- Flowers, J. H., & Wilcox, N. (1982). The effect of flanking context on visual classification: The joint contribution of interactions at different processing levels. *Perception & Psychophysics*, 32(6), 581-591. doi: 10.3758/BF03204214
- Friedman, N. P., & Miyake, A. (2004). The relations among inhibition and interference control functions: a latent-variable analysis. *Journal of Experimental Psychology: General*, 133(1), 101-135. doi: 10.1037/0096-3445.133.1.101
- Gorlyn, M., Keilp, J. G., Tryon, W. W., & Mann, J. J. (2005). Performance test correlates of component factors of impulsiveness. *Personality and Individual Differences*, 38(7), 1549-1559. doi: 10.1016/j.paid.2004.09.014
- Janis, I. B., & Nock, M. K. (2009). Are self-injurers impulsive? Results from two behavioral laboratory studies. *Psychiatry Research*, 169(3), 261-267. doi: 10.1016/j.psychres.2008.06.041
- Keilp, J. G., Sackeim, H. A., & Mann, J. J. (2005). Correlates of trait impulsiveness in performance measures and neuropsychological tests. *Psychiatry Research*, 135(3), 191-201. doi: 10.1016/j.psychres.2005.03.006
- Kirby, K. N., & Maraković, N. N. (1996). Delay-discounting probabilistic rewards: Rates decrease as amounts increase. *Psychonomic Bulletin & Review*, 3(1), 100-104. doi: 10.3758/BF03210748
- Kornblum, S., Hasbroucq, T., & Osman, A. (1990). Dimensional overlap: cognitive basis for stimulus-response compatibility--a model and taxonomy. *Psychological Review*, 97(2), 253-270. doi: 10.1037/0033-295X.97.2.253
- Lange, F., & Eggert, F. (2014). Sweet delusion. Glucose drinks fail to counteract ego depletion. *Appetite*, 75, 54-63. doi: 10.1016/j.appet.2013.12.02

- Lange, F., Seer, C., Rapior, M., Rose, J., & Eggert, F. (2014). Turn It All You Want: Still No Effect of Sugar Consumption on Ego Depletion. *Journal of European Psychology Students*, 5(3), 1-8. doi: 10.5334/jeps.cc
- Logan, G. D., & Cowan, W. B. (1984). On the ability to inhibit thought and action: A theory of an act of control. *Psychological Review*, 91(3), 295-327. doi: 10.1037/0033-295X.91.3.295
- Logan, G. D., Schachar, R. J., & Tannock, R. (1997). Impulsivity and inhibitory control. *Psychological Science*, 8(1), 60-64. doi: 10.1111/j.1467-9280.1997.tb00545.x
- Logue, A. W., & King, G. R. (1991). Self-control and impulsiveness in adult humans when food is the reinforcer. *Appetite*, 17(2), 105-120. doi: 10.1016/0195-6663(91)90066-2
- Mathôt, S., Schreij, D., & Theeuwes, J. (2012). OpenSesame: An open-source, graphical experiment builder for the social sciences. *Behavior Research Methods*, 1-11. doi: 10.3758/s13428-011-0168-7
- Mazur, J. E. (1988). Estimation of indifference points with an adjusting-delay procedure. *Journal of the Experimental Analysis of Behavior*, 49(1), 37-47. doi: 10.1901/jeab.1988.49-37
- Meng, X. L., Rosenthal, R., & Rubin, D. B. (1992). Comparing correlated correlation coefficients. *Psychological Bulletin*, 111(1), 172-175. doi: 10.1037/0033-2909.111.1.172
- Metcalf, J., & Mischel, W. (1999). A hot/cool-system analysis of delay of gratification: dynamics of willpower. *Psychological Review*, 106(1), 3-19. doi: 10.1037/0033-295X.106.1.3
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to

- complex “frontal lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49-100. doi: 10.1006/cogp.1999.0734
- Mobini, S., Grant, A., Kass, A. E., & Yeomans, M. R. (2007). Relationships between functional and dysfunctional impulsivity, delay discounting and cognitive distortions. *Personality and Individual Differences*, 43(6), 1517-1528. doi: 10.1016/j.paid.2007.04.009
- Muraven, M., & Baumeister, R. F. (2000). Self-regulation and depletion of limited resources: Does self-control resemble a muscle? *Psychological Bulletin*, 126(2), 247-259. doi: 10.1037/0033-2909.126.2.247
- Nee, D. E., Wager, T. D., & Jonides, J. (2007). Interference resolution: insights from a meta-analysis of neuroimaging tasks. *Cognitive, Affective, & Behavioral Neuroscience*, 7(1), 1-17. doi: 10.3758/CABN.7.1.1
- Nigg, J. T. (2000). On inhibition/disinhibition in developmental psychopathology: views from cognitive and personality psychology and a working inhibition taxonomy. *Psychological Bulletin*, 126(2), 220-246. doi: 10.1037//0033-2909.126.2.220
- Patton, J. H., & Stanford, M. S. (1995). Factor structure of the Barratt impulsiveness scale. *Journal of Clinical Psychology*, 51(6), 768-774. doi: 10.1002/1097-4679(199511)51:6<768::AID-JCLP2270510607>3.0.CO;2-1
- Posner, M. I., & Rothbart, M. K. (1998). Attention, self-regulation and consciousness. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 353, 1915-1927. doi:10.1098/rstb.1998.0344
- Preuss, U. W., Rujescu, D., Giegling, I., Watzke, S., Koller, G., Zetzsche, T., ... & Möller, H. J. (2007). Psychometrische Evaluation der deutschsprachigen Version der Barratt-Impulsiveness-Skala. *Der Nervenarzt*, 79(3), 305-319. doi: 10.1007/s00115-007-2360-7

- Rachlin, H., & Green, L. (1972). Commitment, choice and self-control. *Journal of the Experimental Analysis of Behavior*, 17(1), 15-22. doi: 10.1901/jeab.1972.17-15
- Rachlin, H., Raineri, A., & Cross, D. (1991). Subjective probability and delay. *Journal of the Experimental Analysis of Behavior*, 55(2), 233-244. doi: 10.1901/jeab.1991.55-233
- Reynolds, B., Ortengren, A., Richards, J. B., & de Wit, H. (2006). Dimensions of impulsive behavior: Personality and behavioral measures. *Personality and Individual Differences*, 40(2), 305-315. doi: 10.1016/j.paid.2005.03.024
- Sharma, L., Kohl, K., Morgan, T. A., & Clark, L. A. (2013). "Impulsivity": Relations between self-report and behavior. *Journal of Personality and Social Psychology*, 104(3), 559-575. doi: 10.1037/a0031181
- Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6), 643-662. doi:10.1037/h0054651
- Stuss, D. T., Murphy, K. J., Binns, M. A., & Alexander, M. P. (2003). Staying on the job: the frontal lobes control individual performance variability. *Brain*, 126(11), 2363-2380. doi: 10.1093/brain/awg237
- Swann, A. C., Bjork, J. M., Moeller, F. G., & Dougherty, D. M. (2002). Two models of impulsivity: relationship to personality traits and psychopathology. *Biological psychiatry*, 51(12), 988-994. doi: 10.1016/S0006-3223(01)01357-9
- Tabibnia, G., Monterosso, J. R., Baicy, K., Aron, A. R., Poldrack, R. A., Chakrapani, S., ... & London, E. D. (2011). Different forms of self-control share a neurocognitive substrate. *The Journal of Neuroscience*, 31(13), 4805-4810. doi: 10.1523/JNEUROSCI.2859-10.2011
- Tangney, J. P., Baumeister, R. F., & Boone, A. L. (2004). High self-control predicts good adjustment, less pathology, better grades, and interpersonal success. *Journal of Personality*, 72(2), 271-324. doi: 10.1111/j.0022-3506.2004.00263.x

Weller, R. E., Cook, E. W., Avsar, K. B., & Cox, J. E. (2008). Obese women show greater delay discounting than healthy-weight women. *Appetite*, 51(3), 563-569.

10.1016/j.appet.2008.04.010

Whiteside, S. P., & Lynam, D. R. (2001). The five factor model and impulsivity: Using a structural model of personality to understand impulsivity. *Personality and Individual Differences*, 30(4), 669-689. doi: 10.1016/S0191-8869(00)00064-7

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Table 1.

Summary statistics and intercorrelations between behavioral and psychometric measures of impulsiveness in study 1 (N = 70)

	Measure	M	SD	Behavioral measures			BIS scales		
				1	2	3	4	5	6
1	DD_log[k]	-7.24	1.42	.-					
2	GNG_number of false alarms	7.55	5.82	.12	-				
3	GNG_coefficient of variation	23.64	4.79	.10	.04	-			
4	BIS_total	59.52	10.18	.10	.36**	.16	-		
5	BIS_attentional impulsiveness	15.67	3.48	.07	.41***	.20	.71***	-	
6	BIS_motor impulsiveness	21.39	4.34	.09	.27*	.15	.90***	.46***	-
7	BIS_non-planning impulsiveness	22.46	4.39	.07	.23	.07	.87***	.38**	.72***

Note: DD = delay discounting, GNG = go/no-go task, BIS = Barratt Impulsiveness Scale (sum scores)

* $p < .05$, ** $p < .01$, *** $p < .001$

Table 2.

Summary statistics and intercorrelations between behavioral and psychometric measures of impulsiveness in study 2 (N = 115)

	Measure	M	SD	Behavioral measures		BIS scales		
				1	2	3	4	5
1	GNG_number of false alarms	5.61	4.69	-				
2	GNG_coefficient of variation	20.25	7.05	.17	-			
3	BIS_total	60.30	8.94	.12	-.04	-		
4	BIS_attentional impulsiveness	15.76	2.84	.07	.05	.72***	-	
5	BIS_motor impulsiveness	21.93	3.99	.08	-.08	.87***	.51***	-
6	BIS_non-planning impulsiveness	22.61	4.08	.13	-.04	.84***	.39***	.57***

Note: DD = delay discounting, GNG = go/no-go task, BIS = Barratt Impulsiveness Scale (sum scores)

* $p < .05$, ** $p < .01$, *** $p < .001$

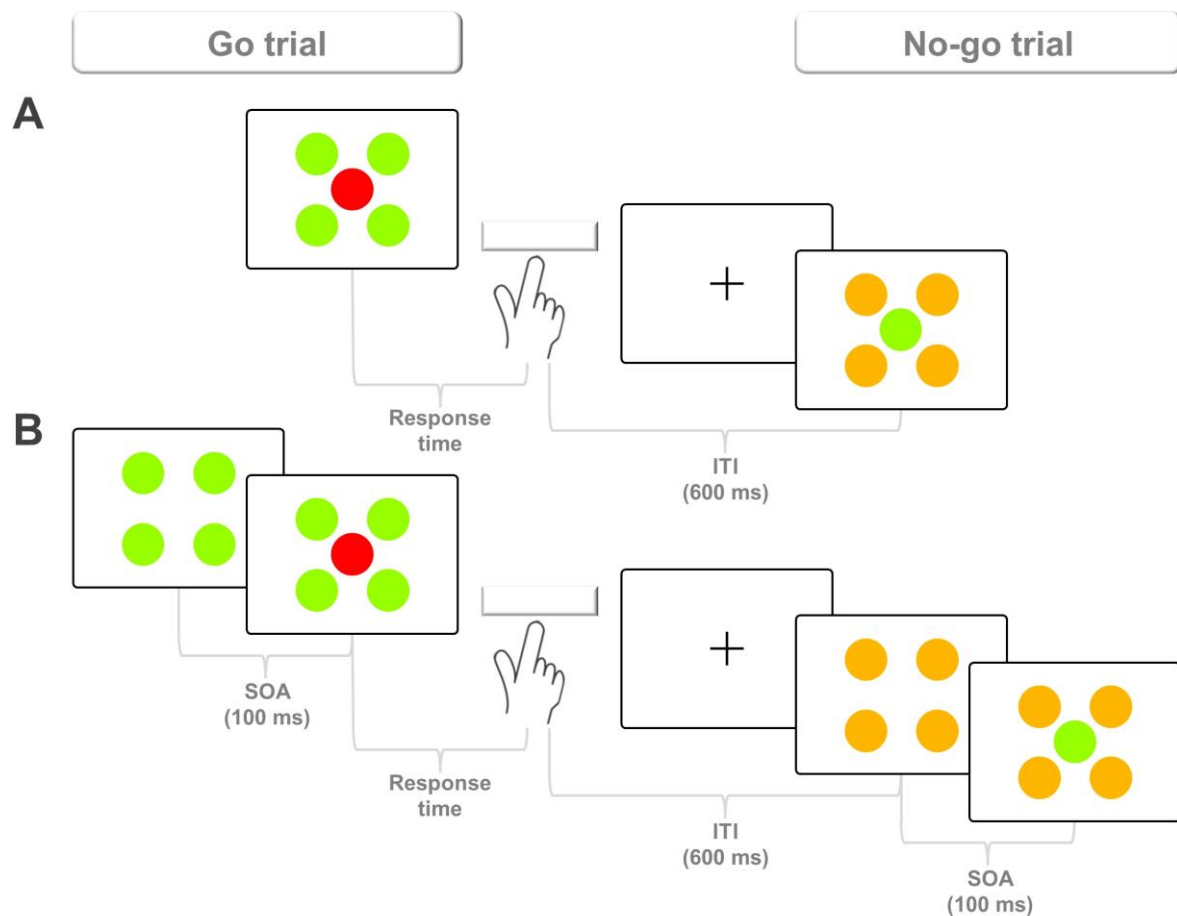


Figure 1. Task sequences of the go/no-go tasks administered in Study 1 (A) and Study 2 (B). Participants were required to press the space bar as fast as possible whenever a frequent go stimulus (red circle) appeared in the center of the computer screen, but to withhold responses to infrequent no-go stimuli (green circle) for 1500 ms. Responses were followed by an inter-trial interval (ITI) of 600 ms. Four task-irrelevant distractors (circles randomly varying in color) were configured around the central target stimulus. Distractor and target stimuli appeared simultaneously on the screen in Study 1 (A). In Study 2 (B), a stimulus onset asynchrony (SOA) of 100 ms was introduced to alter the balance between facilitative and interfering priming processes.